

Technical Datasheet - VBF-T2R2-DSH-01

Case t2_r2 - Virtual Battery Factory - generated 2026-08-26

Technical Datasheet - all values mechanically sourced (see source column)

Field	Value	Source
Rated capacity	5.0 Ah (nominal); 6.9089 Ah (1C DFN simulated, 25 C)	chen2020_dump.json / final_1c_dfn.json
Nominal voltage / window	3.8751 V midpoint; 2.5-4.2 V	final_calc.json / chen2020_dump.json
Rated energy	25.3513 Wh	final_calc.json (time integration of V x I)
Energy density	522.4524 Wh/kg (869.1832 Wh/L)	final_calc.json; contract caliber - electrolyte excluded: ■■■■■■■■■■■■■■■■■■■■■
Max continuous discharge	1C verified 6.9089 Ah, T_max 305.20 K	final_1c_dfn.json
Fast-charge capability	4C @45 C: NO plating (min anode potential +0.0410 V); T_max 360.09 K = 86.9 degC; accepts 0.958 Ah before 4.2 V cutoff (~13.9 % - honestly reported)	final_4c_dfn.json
Operating temperature range	verified endpoints: -20 C (retention 99.62 %) to +45 C (4C charge); beyond endpoints not simulated	final_lowt_spme.json / final_4c_dfn.json (honest scope)
Cycle life (SEI-growth, task-caliber)	99.9 nm @100 cyc (PASS <= 500); 330.1 nm @500 cyc (PASS <= 550)	final_aging100.json / final_aging500.json
Cycle life (EOL cycle count)	Not simulated: aging reports SEI thickness only; trajectory 0.547 -> 0.531 Ah not an EOL basis (annotation, not fabricated)	final_aging500.json
Safety determination	4C plating: PASS (>0 V); T_max 86.9 degC at 4C, no task red-line exceeded; no other abuse protocols run	final_4c_dfn.json
Dimensions and mass	65 x 1580 x 284 um; 48.524 g contract caliber / 61.033 g incl. electrolyte (annotation)	geometry dump + final_calc.json + pore-volume calc